7.17 Work Near a Rail System Grade Crossing

Purpose:

This layout shows a typical setup for a single lane closure on a two-lane, two-way highway where due to the proximity to a railway crossing, extra care should be taken to minimize the probability of conditions being created by:

1) lane restrictions,
2) flagging operations, and/or
3) other operations where vehicles may get stopped within the grade crossing.

Avoid stopping any vehicles within 5.0 metres on approaches to and departures from crossings, measured from either the closest or farthest rail.

Standard:

- Before work begins, users of this layout shall coordinate with the railroad company, or light rail transit agency operating the line. Getting approvals near rail could be a lengthy process. Therefore, early coordination with rail/transit organizations is required.
- Rail and transit organizations have specific rules for those working in the vicinity of their tracks. Therefore, their rules, and those of Transport Canada will have to be adhered to.
- TCP's shall determine the potential for upstream devices such as traffic signals leading to queuing of vehicles across active rail tracks. Coordination between TCP's to determine upstream storage capacity will help regulate the numbers of vehicles released per direction to mitigate queues stopping on tracks. The R-285 "KEEP TRACKS CLEAR" may be added.
- If the queuing of vehicles across active rail tracks cannot be avoided due to reasons such as the location of upstream traffic signals, an additional flagger shall be provided at the grade crossing to prevent vehicles from stopping within the grade crossing.
- TCP's shall obey the Railway Signals.
- When used at night, the TCP station shall be illuminated with overhead lighting.

Guidance:

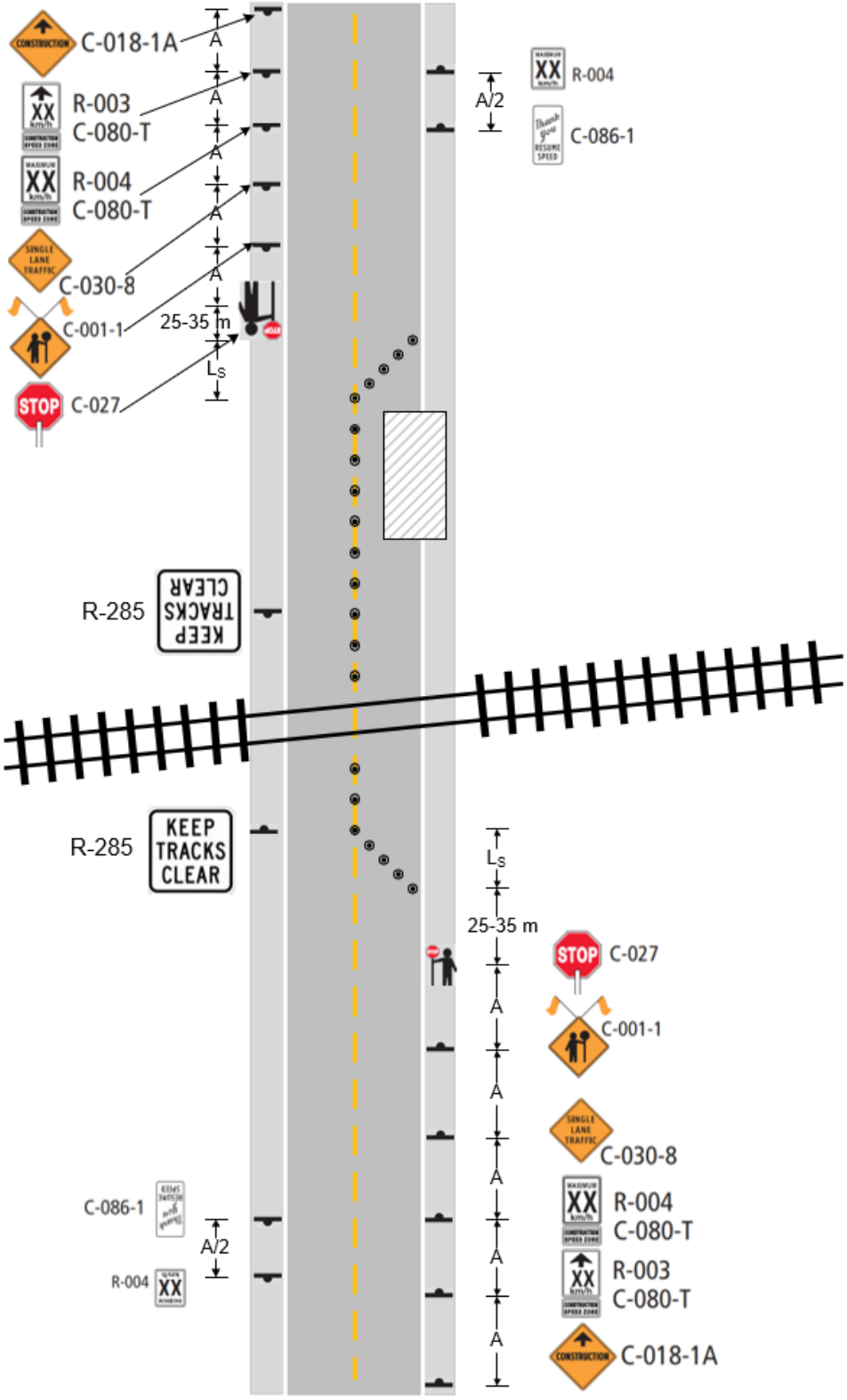
- Extending the buffer space (on the lane closure side) on the upstream side of the grade crossing will help prevent the backup of traffic across the railway crossing created by the TCP operation.
- Channelizing devices shall not be placed within 2.5 metres on either side of the closest and farthest rail.
- The distance between the TCP and the Traffic Control Person Ahead C-001-1 sign should not exceed 150 metres.
- Where Crew Working – Maximum Speed C-002-2 signs establish a Temporary Speed Zone, the C-002-2 should be placed upstream of the C-004 or C-018-1. - Thank You Resume Speed C-086-1 signs should be placed across from the Crew Working – Maximum Speed C-002-2 signs in the opposing lanes.

Options:

- An additional Traffic Control Person Ahead C-001-1 sign may be added to the far side of the road to provide queued drivers with increased awareness of the TCP position.
- The Flagger Ahead C-001-2 sign or Prepare to Stop C-029 sign may be used for additional advance warning where TCPs are stopping traffic.
- A Prepare to Stop C-029 sign may replace the Single Lane Traffic C-030-8 sign for other applications that require traffic to stop (e.g., equipment crossing road).
- Run-in delineation, LR may be omitted during period where queues are low, permitting the full release of the queue in each direction.

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Figure 7.17: Work Near a Rail System Grade Crossing



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PART B – TRAFFIC CONTROL

Section 8: Traffic Control Layouts – Multilane Undivided Roadways

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PART B – TRAFFIC CONTROL

Section 8: Traffic Control Layouts – Multilane Undivided Roadways

LEGEND			
	Flexible Drum		Paint Truck
	Tubular Marker		Escort Truck
	Cone		Chaser Vehicle
	Sign		Vehicle Mounted Rear Crash Attenuator
	Traffic Control Person		
	Work Activity Area		360° Flashing Light
	Work Truck		Portable Traffic Signal
	Shadow Vehicle		Barricade and Fencing
	Shadow Vehicle #1		Flashing Arrow Board (FAB)
	Shadow Vehicle #2		Flashing Arrow Board (FAB) in caution mode
	Buffer Vehicle		Dynamic Message Sign (DMS)

TABLE A – TAPER LENGTHS

Taper Types (m)		Regular Posted Speed Limit (km/h)							
		≤50	60	70	80	90	100	110	120
Merge Taper Length	L_M	35	55	160	190	210	230	250	280
Lane Shift Taper Length	L_L	30	50	80	100	110	120	130	140
Downstream Taper Length	L_D	30	30	30	30	30	30	30	30
TCP, Signal, and Shoulder Taper Length (min. 5 devices)	L_S	5	8	15	15	15	15	15	15
Minimum Tangent Length between Tapers	L_T	30	60	160	190	210	230	250	280
Run-In Length on Centreline	L_R	40	50	60	60	70	80	90	100

TABLE B – DEVICE SPACING LENGTHS

Device Spacing (m)		Regular Posted Speed Limit (km/h)							
		≤50	60	70	80	90	100	110	120
Construction Sign Spacing	A	40	60	80	100	150	150	200	200
Buffer Space	B	30	40	60	80	110	140	170	200
Roll-Ahead Buffer Distance	R	30	30	40	40	40	50	50	50
Channelizing Device Spacing for Tapers	C	10	10	15	15	15	15	15	15
Channelizing Device Spacing on Curves and Tangents	D	10	10	30	30	40	40	40	50

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8.1 General Information– Multilane Undivided Roadway

The traffic control layouts are considered the minimum standard. The associated text description of each figure highlights the key standards as well as guidance and options that can be considered by the user. The following information is typical on most layouts in this section:

Standard:

- For short-duration work, a Crew Working Ahead C-004 sign shall be used.
- For long-duration work, a Construction Ahead C-018-1A sign shall be used.
- Where the speed limit is ≥ 70 km/h, a buffer space shall be used.
- A flashing arrow board (FAB) shall be used when a lane is closed.
- All work, buffer, and shadow vehicles shall be equipped with a 360-degree flashing light and 4-way flashers.

Guidance:

- Where the speed limit is ≥ 70 km/h a buffer vehicle should be used when workers are present.
- A vehicle-mounted crash attenuator is recommended for use on buffer vehicles.

Options:

- Where cyclists are regularly observed using the shoulder, a Share the Road W-132-1 sign may be used with an appropriate cycling hazard tab (see *Section 18: Traffic Control Layouts – Bicycle Lanes* for additional information).
- A portable dynamic message sign (DMS) may be used to provide advance messaging for drivers.
- In low speed (≤ 60 km/h) urban areas (within municipal boundaries), customization of traffic control layouts, including closer device spacing and shorter taper lengths, may be necessary to maintain access and mobility. In these cases, document why adjustments are being made.

8.2 Typical Construction Speed Zone Signing – Multilane Undivided Roadway

Purpose:

Construction speed limits are regulatory speeds established in Construction Speed Zones within long-term construction and maintenance project areas where there are continuous hazards for motorists or where workers are in close proximity to active travel lanes.

Construction Speed Zones should be applied prudently because **overuse reduces effectiveness.** Drivers should be able to perceive the need to reduce speed.

Standard:

- Conflicting speed limit signs within the Construction Speed Zone shall be covered or removed.
- The same Construction Speed Zone signing is required in opposing directions.

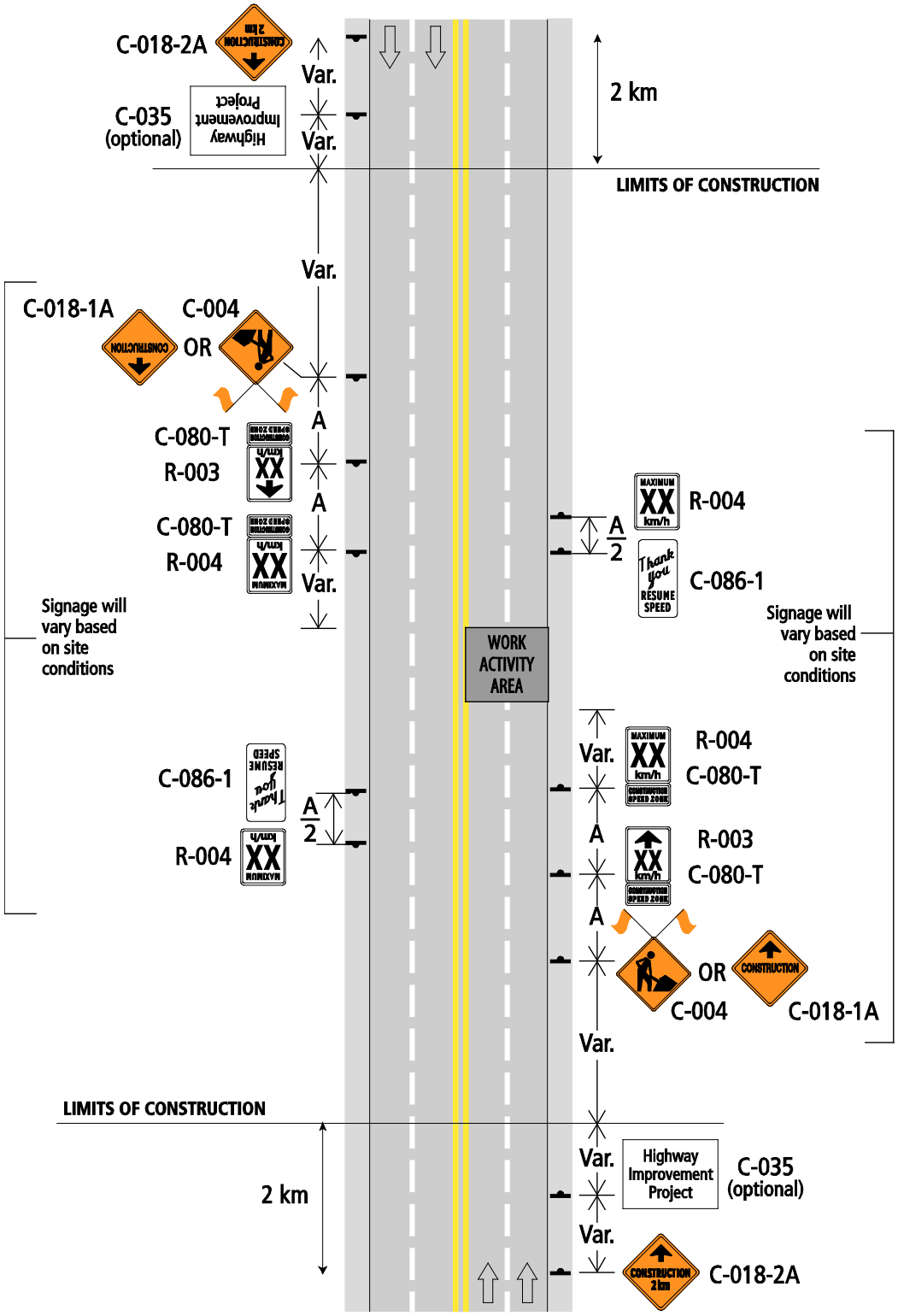
Guidance:

- The Maximum Speed R-004 speed sign with a Construction Speed Zone C-080-T distance tab should be positioned at the beginning of the active work area.
- Construction speed limit signs should be covered or removed when no work is occurring and other hazards are not present.
- It is important to record when construction speed limits are installed and covered/removed.
- See also *Section 2.4: Management of Speed* for information on using Construction Speed Zones.

Options:

- Use of the Construction Project C-035 sign is project-dependent. It is typically used only for large projects. See *Appendix B: Standard Construction Signs* for information on using this sign.
- If secondary signs are applied, they may be positioned based on:
  - If the median is more than 2 metres wide, secondary signs may be placed in the median.
  - If the median is 2 metres wide or less, secondary signs may be placed on the opposite shoulder.

Figure 8.2: Typical Construction Speed Zone Signing – Multilane Undivided Roadway



8.3 Emergent Work (<5 Minutes) – Multilane Undivided Roadway

Purpose:

Emergent work involves very short-duration activities for which setting up and taking down temporary traffic control devices may take more time than the actual work, and expose workers to greater risk. Each entry onto the travelled portion of the roadway lasts less than 1 minute, and the total time to complete the task is less than 5 minutes.

This does not include emergencies, which are situations which require immediate response to save lives or prevent serious injury using whatever resources are available.

It may occur when an unanticipated situation or event—one that presents a risk to the travelling public—is discovered during travel or work activities. It can be considered unplanned, urgent maintenance work. A Traffic Control Plan is not required for emergent work.

Emergent work may include the removal of debris from the roadway (e.g., tree limbs, lost cargo, dead animals, tire and other vehicle debris, and the manual removal of rock). Isolated pothole patching—patching 1 or 2 potholes in a 1-kilometre section of road—may be considered emergent or brief-duration work. However, it is mobile work when the work crew has to stop several times in succession within a 1-kilometre section to patch multiple potholes.

Standard:

- A risk evaluation is required to determine whether or not the work activity is considered emergent (see *Table C – Risk Evaluation for Emergent or Brief-Duration Work* in *Section 6.10* or *Appendix F*).
- The entire work activity shall be completed in less than 5 minutes.
- The work vehicle shall use a flashing arrow board (FAB) in caution mode or a 360-degree flashing light and 4-way flashers.

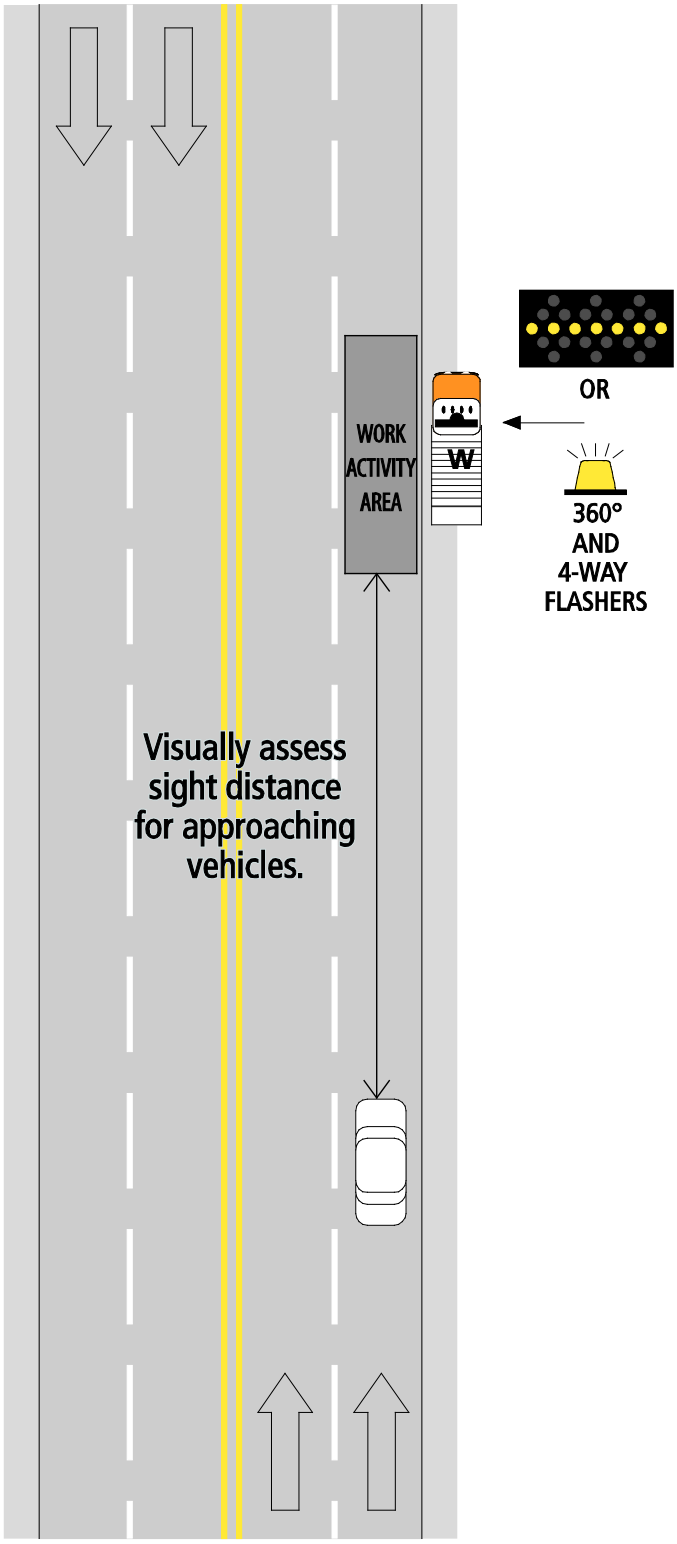
Guidance:

- If a dynamic message sign (DMS) or flashing arrow board (FAB) is used for an operation on the shoulder, it should be set to caution mode.
- The position of the work vehicle in relation to the work area may be adjusted based on the available sight distance, shoulder/off-roadway conditions, and work being performed.

Options:

- None at this time.

Figure 8.3: Emergent Work – Multilane Undivided Roadway



8.4 Brief-Duration Work (<15 Minutes) – Multilane Undivided Roadway

Purpose:

Brief-duration work is generally planned, although the exact location or extent of the work required may not be fully known. It requires less than 15 minutes to complete.

Brief-duration work may include:

- locating drainage structures or other roadway features or components
- cleanup of material spills and removing debris from the roadway (e.g., small fallen trees, larger tree limbs, crash debris, etc.)
- quick repairs intended as a partial or temporary response to damage or failure

If the work is expected to last more than 15 minutes, additional resources will be needed to implement a short-duration work zone. The work should be delayed until the appropriate work zone equipment and devices are available.

Isolated pothole patching—patching 1 or 2 potholes in a 1-kilometre section of road—may be considered emergent or brief-duration work. However, it is mobile work when the work crew has to stop several times in succession within a 1-kilometre section to patch multiple potholes.

Standard:

- A Crew Working Ahead C-004 is required in advance of the work.
- A risk evaluation is required to determine whether or not the work activity is considered brief (see *Table C – Risk Evaluation for Emergent or Brief-Duration Work* in *Section 6.10* or *Appendix F*).
- The work activity shall be completed in less than 15 minutes.
- The work vehicle shall use a flashing arrow board (FAB) in caution mode or a 360-degree flashing light and 4-way flashers.

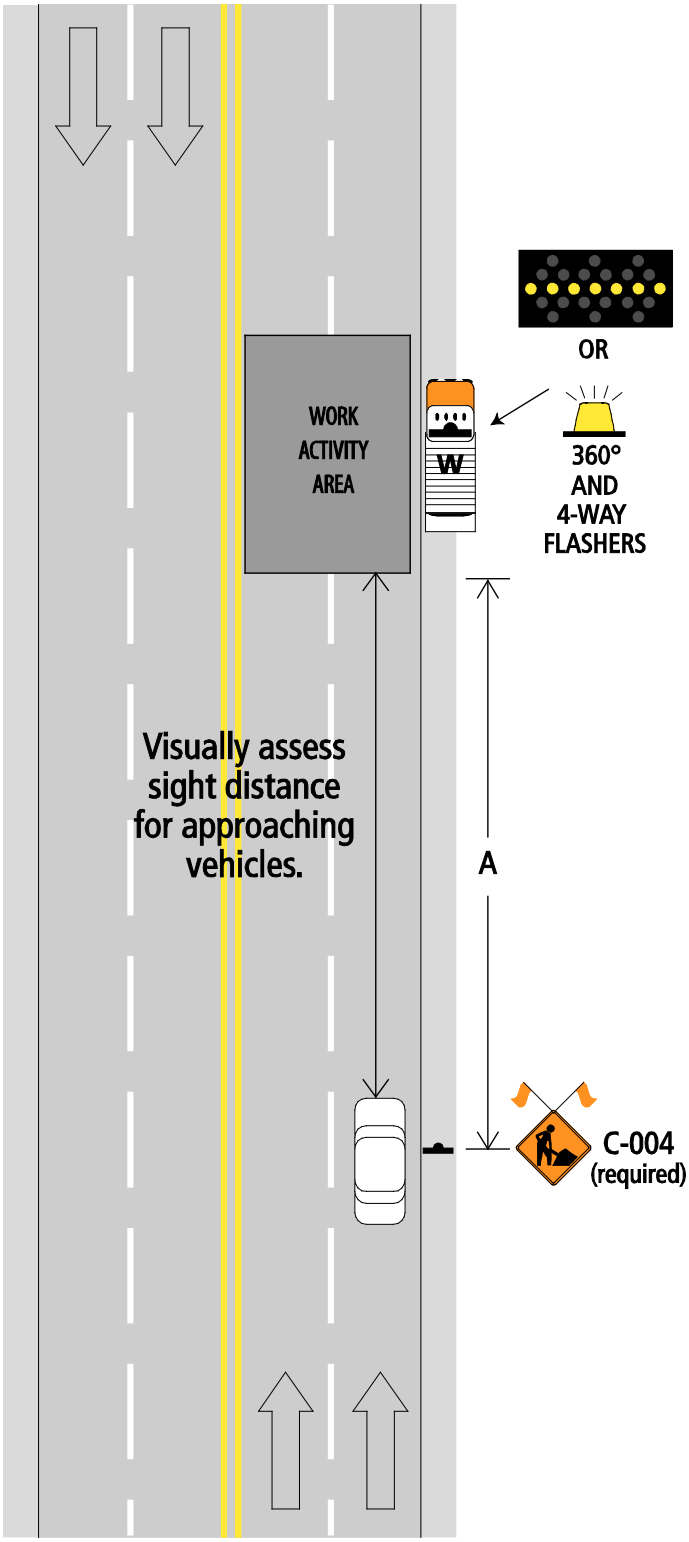
Guidance:

- If a dynamic message sign (DMS) or flashing arrow board (FAB) is used for an operation on the shoulder, it should be set to caution mode.
- The position of the work vehicle in relation to the work area may be adjusted based on the available sight distance, shoulder/off-roadway conditions, and the work being performed.

Options:

- None at this time.

Figure 8.4: Brief-Duration Work – Multilane Undivided Roadway



8.5 Work on Shoulder – Short and Long Duration

Purpose:

Stationary work on the shoulder takes place outside the travel lanes but occupies part or all of the shoulder area.

If shoulder work encroaches into a travel lane, a full lane closure shall be implemented.

Standard:

- When work is in progress and workers are present, a flashing arrow board (FAB) in caution mode is required in the taper or immediately in advance of the work area.
- A barricade is required on the approach side of the work activity area for long-duration work.

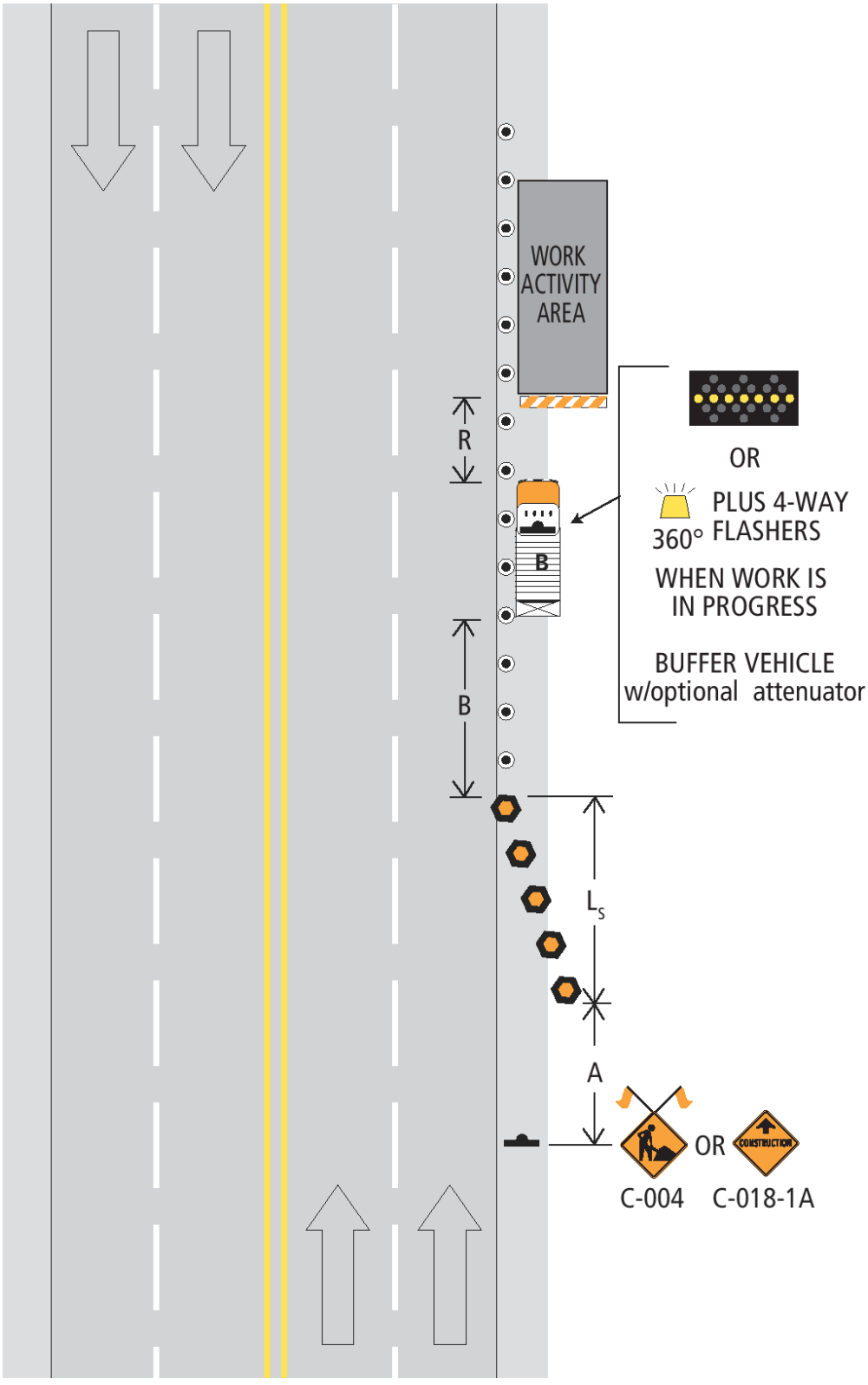
Guidance:

- When work is not in progress but the work area has not been cleared, care should be taken to isolate it from the travelled roadway

Options:

- Advance warning signs may be applied in the opposing direction of travel.
- If the speed limit is ≤ 60 km/h, or there is insufficient room, tubular markers may replace drums for the taper.
- The channelizing devices used alongside the work area may be tubular markers or cones.
- Advance warning signs may be omitted for short-duration work if the shadow vehicle displays a vehicle-mounted dynamic message sign (DMS) or flashing arrow board (FAB) in caution mode and uses a 360-degree flashing light and 4-way flashers.

Figure 8.5: Work on Shoulder – Short and Long Duration



8.6 Right Lane Closed – Short and Long Duration

Purpose:

This layout shows the typical setup for closure of the right travel lane on a multilane undivided highway.

Standard:

- A Right Lane Closed Ahead C-130-R sign with a C-130-T distance tab and a second Right Lane Closed Ahead C-130-R sign are required.
- A barricade is required on the approach side of the work activity area for long-duration work.

Guidance:

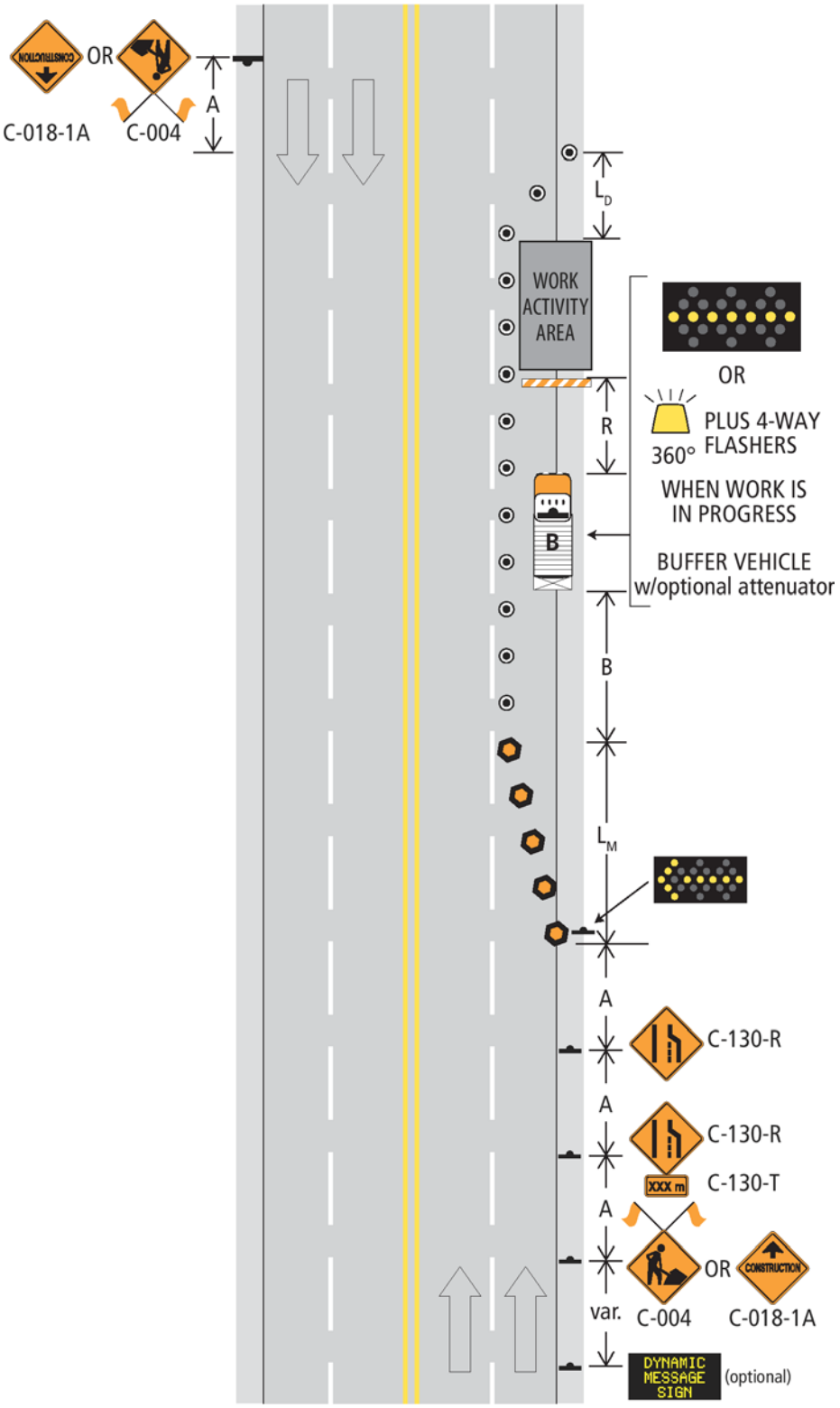
- Where shoulders are wide and may be mistaken for travel lanes, shoulder tapers should be considered and installed in accordance with *Table A – Taper Lengths.*

Options:

- If the speed limit is ≤ 60 km/h:
  - The upstream Right Lane Closed C-130-R sign may be omitted and the Crew Working Ahead C-004 sign or Construction Ahead C-018-1A sign moved downstream by Table B, Distance A.
  - Tubular markers may be used for leading tapers instead of drums.
- The flashing arrow board (FAB) may be replaced as shown below.

Speed Limit	Workers or Work Vehicles Present	No Workers or Vehicles Present
≤ 60 km/h	Lane Closure Arrow C-053 sign plus work vehicle with 360° flashing light and 4-way flashers.	Lane Closure Arrow C-053 sign plus barricade and Type A yellow warning light.
≥ 70 km/h	No substitution.	

Figure 8.6: Right Lane Closed – Short and Long Duration



8.7 Left Lane Closed – Short and Long Duration

Purpose:

This layout shows the typical setup for closure of the left travel lane on a multilane undivided highway.

Standard:

- A Left Lane Closed Ahead C-130-L sign with a C-130-T distance tab and a second Left Lane Closed Ahead C-130-L sign are required.
- A flashing arrow board (FAB) shall be used where the posted speed limit is ≥ 70 km/h.
- Barricades are required at each end of the work activity area for long-duration work.

Guidance:

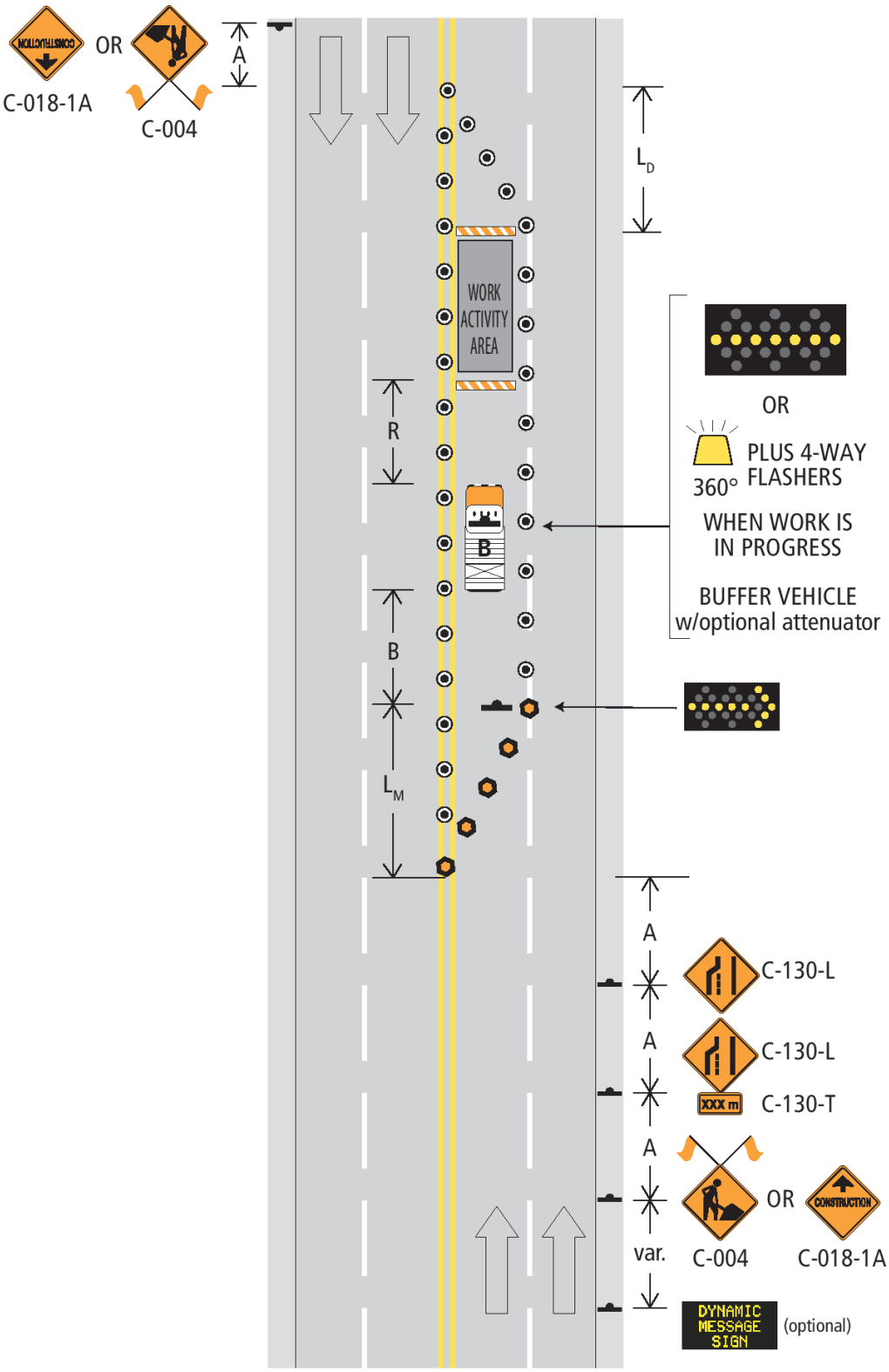
- If adequate space cannot be maintained for worker safety, it may be necessary to close the left lane in the opposing direction.

Options:

- Secondary Left Lane Closed C-130-L signs may be installed on the left side of the highway.
- If the speed limit is ≤ 60 km/h:
  - The upstream Left Lane Closed C-130-L sign may be omitted and the Crew Working Ahead C-004 sign or Construction Ahead C-018-1A sign moved downstream by Table B, Distance A.
  - Tubular markers may be used for leading tapers instead of drums.
- The flashing arrow board (FAB) may be replaced as shown below.

Speed Limit	Workers or Work Vehicles Present	No Workers or Vehicles Present
≤ 60 km/h	Lane Closure Arrow C-053 sign plus 360° flashing light and 4-way flashers.	Lane Closure Arrow C-053 sign plus barricade and Type A yellow warning light.
≥ 70 km/h	No substitution.	

Figure 8.7: Left Lane Closed – Short and Long Duration



8.8 Centre Lane Closure (≤ 60 km/h) – Short and Long Duration

Purpose:

This layout shows the typical setup for a single lane closure of the centre lane on a low-speed multilane roadway.

Standard:

- A Centre Lane Closed Ahead C-030-1A sign with a C-130-T distance tab and a Centre Lane Closed C-030-2 sign are required in advance of the work.
- If there is insufficient room for signs in the centre median, a dynamic message sign (DMS) is required.
- A buffer space shall always be incorporated into the layout.
- A barricade is required on each end of the work activity area for long-duration work.

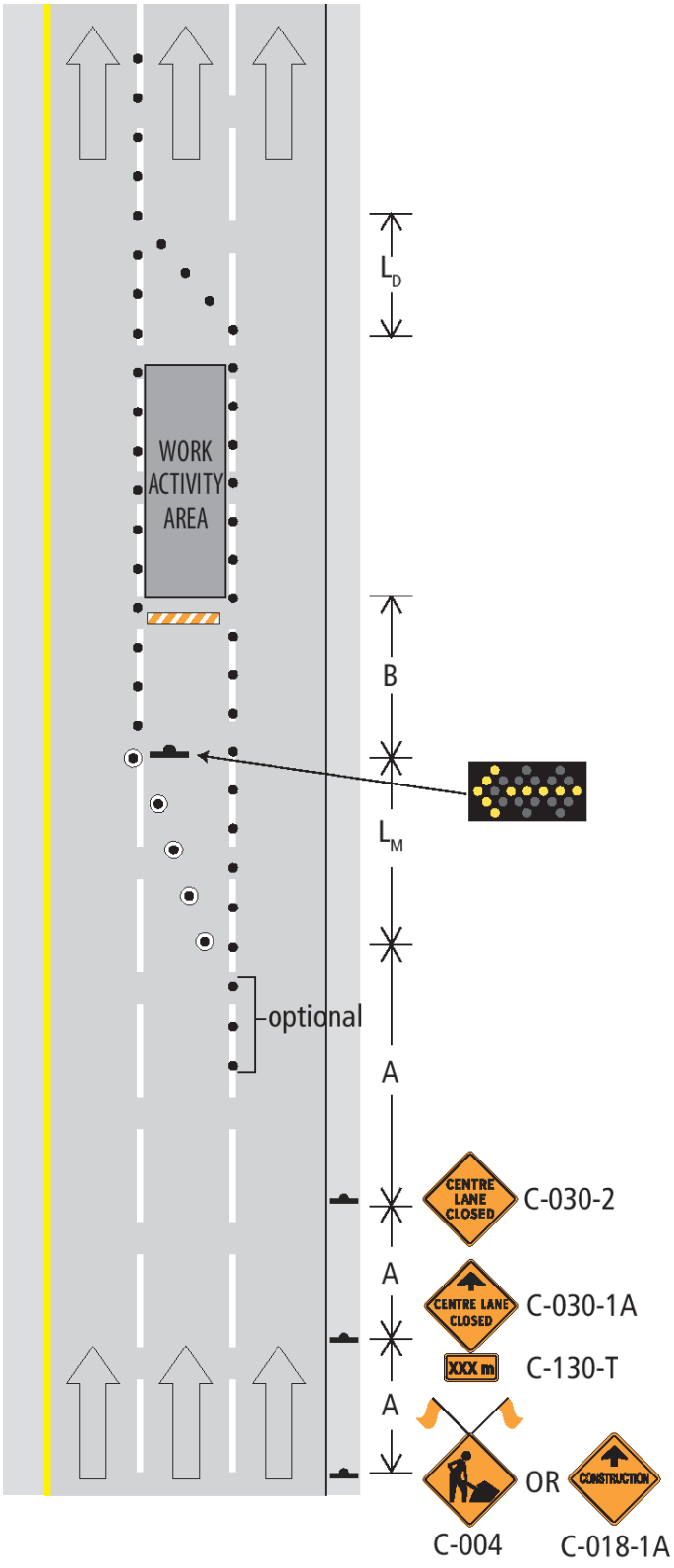
Guidance:

- The spacing of channelizing devices may be reduced to prevent traffic from entering the work area.
- Provided that traffic volumes are sufficiently low, it may be necessary to close two lanes to maintain adequate space for worker safety.
- If space is available, a short, single row of channelizing devices should be added in advance of the traffic split to keep vehicles in their lanes.

Options:

- Interior lane traffic may be directed to either the left or the right lane by using a flashing arrow board (FAB) with the arrow pointing in the direction of the merge taper.

Figure 8.8: Centre Lane Closure (≤ 60 km/h) – Short and Long Duration



8.9 Centre Lane Closure (≥ 70 km/h) – Short and Long Duration

Purpose:

This layout shows the typical setup for a single lane closure of the centre lane on a high-speed multilane roadway.

Standard:

- A Right Lane Closed Ahead C-130-R sign with a C-130-T distance tab and a second Right Lane Closed Ahead C-130-R sign shall be placed in advance of the work.
- A Centre Lane Closed C-030-2 sign is required in advance of the lane shift.
- If there is insufficient room for signs in the centre median, a dynamic message sign (DMS) is required.
- Two flashing arrow boards (FABs) shall be used, one in each taper, as shown in the diagram.
- A barricade is required on the approach side of the work activity area for long-duration work.

Guidance:

- If the alignment is such that the two flashing arrow boards (FABs) create confusion, the minimum tangent length (LT distance) between the end of the merging taper and beginning of the shift taper should be extended so that road users can focus on one flashing arrow board (FAB) at a time.
- The spacing of channelizing devices may be reduced to prevent traffic from entering the work area.
- Provided that traffic volumes are sufficiently low, it may be necessary to close two lanes to maintain adequate space for worker safety.
- A double lane closure (leaving only one lane open) may be implemented if traffic volumes are sufficiently low. A flashing arrow board (FAB) will be required for each closed lane.
- Where shoulders are wide and may be mistaken for travel lanes, shoulder tapers should be considered and installed in accordance with *Table A – Taper Lengths*.

Options:

- Interior lane traffic may be directed to either the left or the right lane by using a flashing arrow board (FAB) with the arrow pointing in the direction of the merge taper.